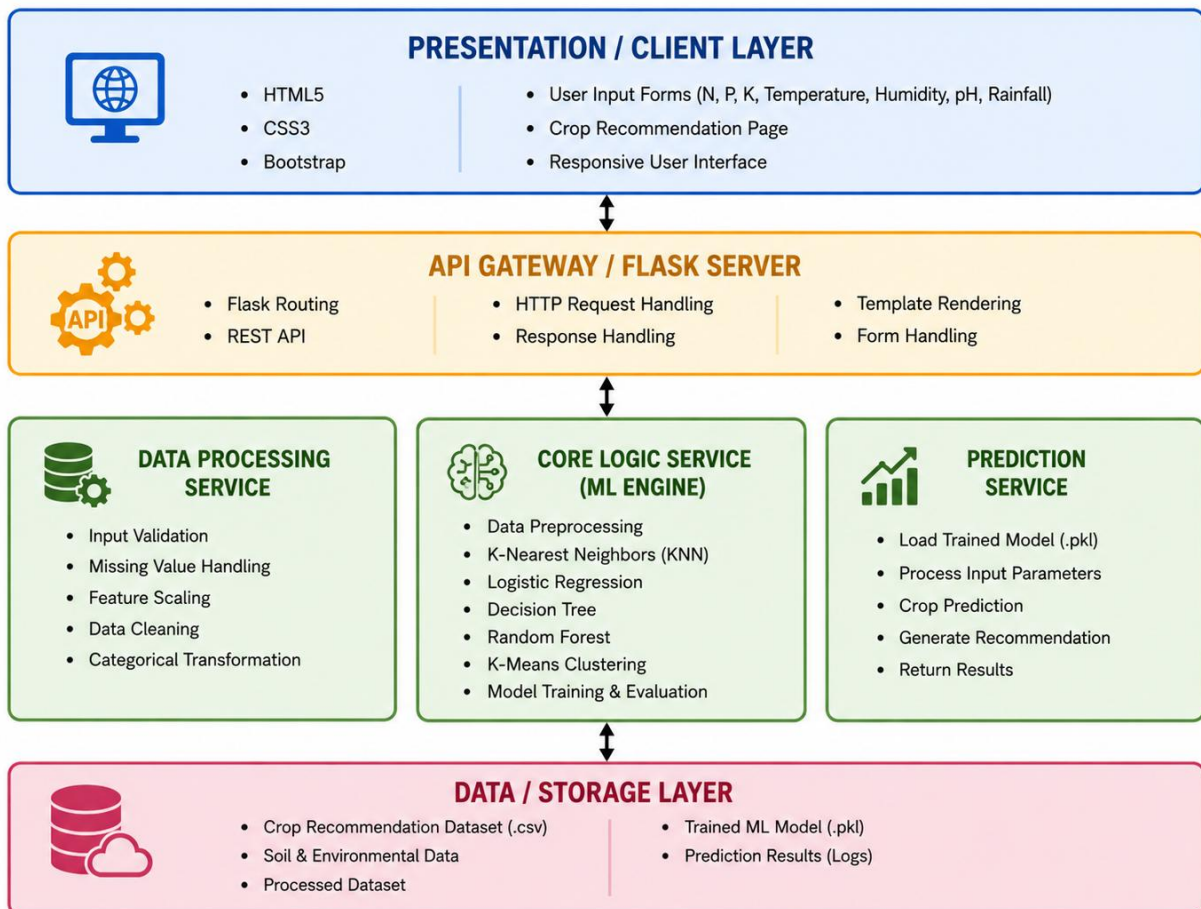


Solution Architecture

Date	15 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Provides the user interface where farmers enter soil and environmental parameters and view crop recommendations.</i>	<i>HTML5, CSS3, Bootstrap, Flask Templates</i>
<i>API Gateway</i>	<i>Handles HTTP requests, routes user requests to the appropriate application logic, and manages communication between the client and server.</i>	<i>Flask Routing</i>
<i>Microservices / Core Logic</i>	<i>Processes user inputs, performs data validation, loads the trained ML model, predicts the most suitable crop, and returns the result.</i>	<i>Python, Flask, Scikit-learn, Pandas, NumPy, Pickle</i>
<i>Database / Storage</i>	<i>Stores the crop recommendation dataset and the trained machine learning model used for predictions.</i>	<i>CSV Dataset, Pickle (.pkl), Local Storage</i>